Welcome to the costume design guild, todays topic is about fashion: from concept to ready to wear garment

Transparency

My master research

- Running these guilds as part of my master research
- Needs to be recorded for documentation

Block B:
- Test runs of “workshop” set-up
- Practise teaching
- **Fashion content**
- In-person only

Block C:
- “Real” runs of my research
- Data gathering
- **Costume content (for games)**
- In-person but also streamed

My master research is about figuring out which costume design principles are important for video game design, and I will find this out by observing how you guys interact with and react to the workshop content I will be preparing the coming blocks
I am running these guilds as part of my masters research and therefore need to be recorded for documentation.

PRE-SESSION QUESTIONS

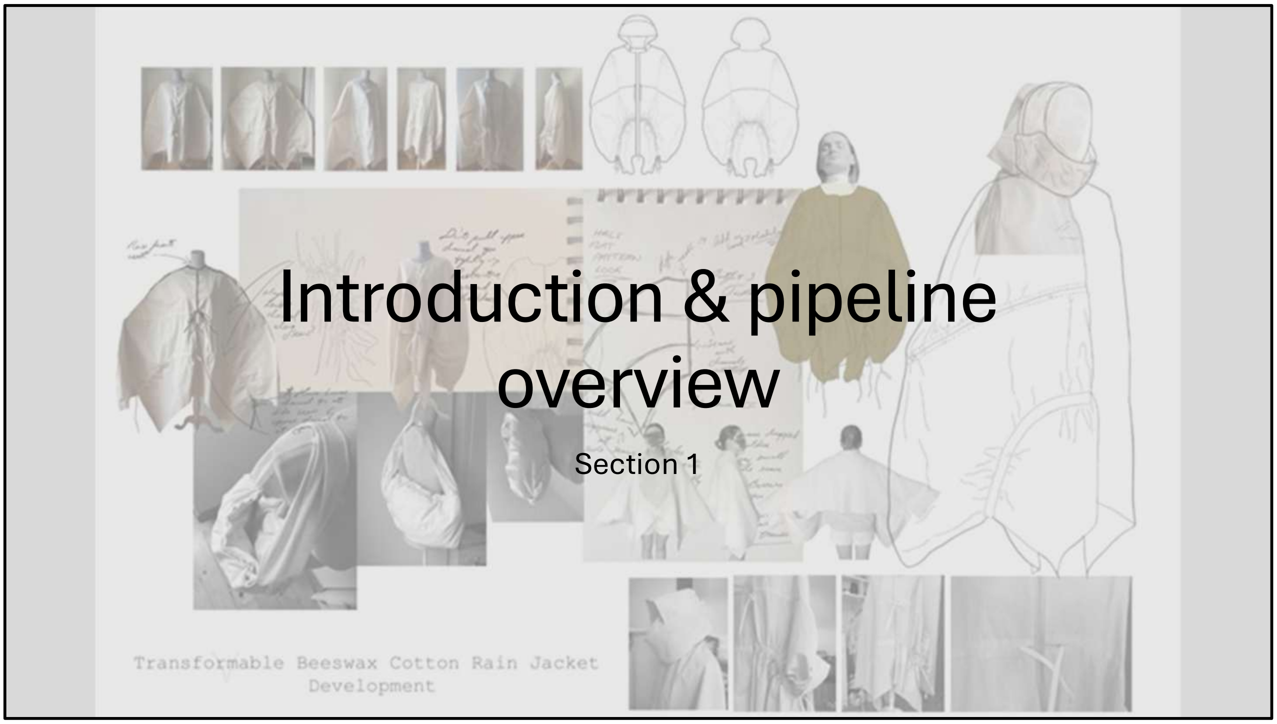
Red:
- No
- No experience
- Unfamiliar
- Not confident

Yellow:
- Maybe
- Some experience
- Somewhat familiar
- Moderately confident

Green:
- Yes
- Experienced
- Very familiar
- Very confident

First I want to ask you a few questions, please answer with the coloured cards so I can collect some data of your knowledge levels

- Have you ever sewn a garment before?
- Do you think you can identify what garment a flat pattern makes just by looking at it?
- Do you know what a basic block is?
- How confident do you feel understanding how flat patterns become 3D garments? (scale)
- Have you used any digital fashion software (CLO3D, Marvelous Designer)?



Lets start with an introduction and the fashion pipeline overview

Image source: https://pin.it/Sid5KjuLz

Introduction

Agenda:

1. Why we’re here
2. The full pipeline
3. Design routes context
4. Patterning deep-dive
5. Activity 1: patterning
6. Prototyping and construction
7. Why this matters for character design
8. Activity 2: 3D shape creation

The agenda of today is…

Source: https://www.artstation.com/artwork/YGoogY
https://www.artstation.com/zahraa97

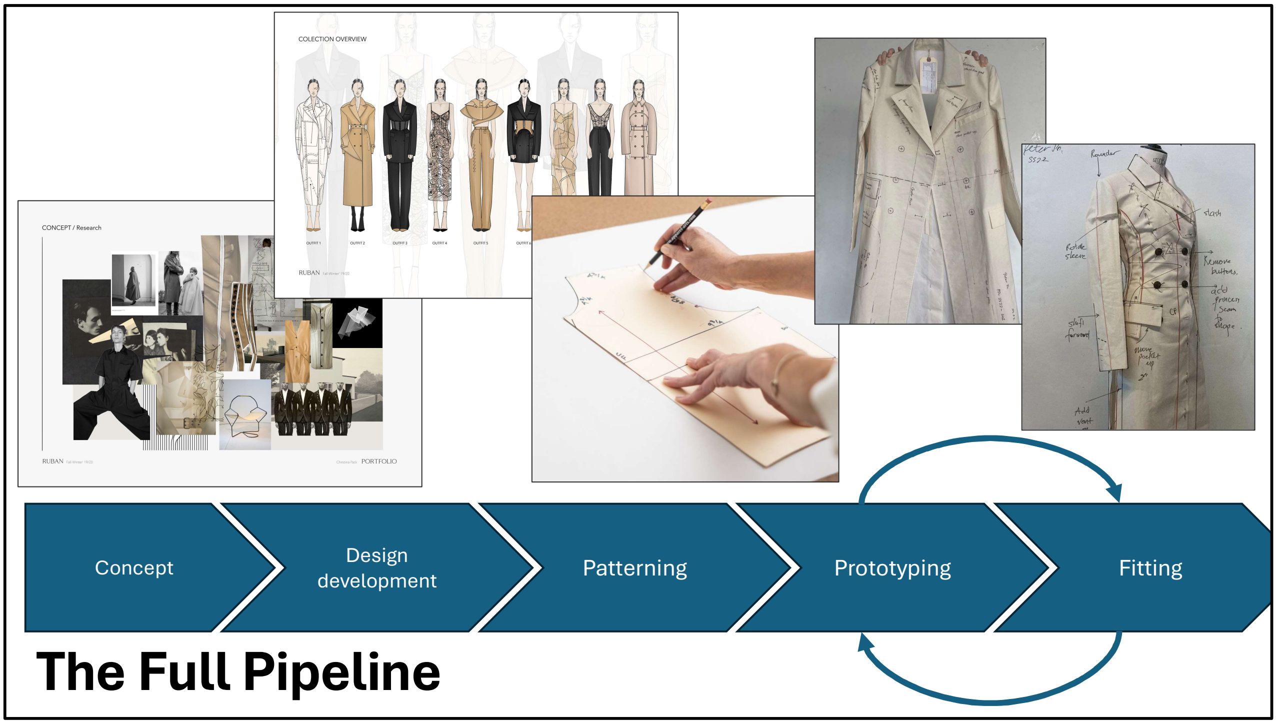
Welcome & Why We're Here

Garments = constructed = follows real-world logic

Understanding this = enhanced believability / realism

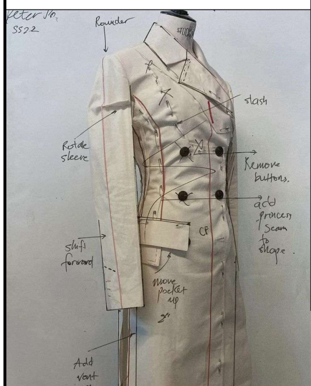
So you're learning character design (or concept for characters) - but how well do you understand the costumes you're designing?
Costume = constructed garment = needs to follow real-world logic
Today: Understanding how flat fabric becomes 3D form

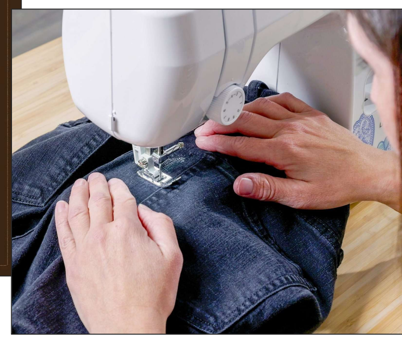
Source: https
://www.artstation.com/artwork/YB0B2Y
https://www.artstation.com/zahraa97



This is a broad overview of the full fashion pipeline, starting with concept in which we collect images and place them artistically/in an inspired way
Then a bunch of sketching, sometimes 100s of outfits drawn before deciding on the key outfits in a collection
Then comes patterning, later I will tell you about the different ways of creating patterns
Then prototyping, here you test the patterns, sew it from cheap cotton
And fitting, where you figure out what needs adjusting to achieve the desired look
You keep returning to making a new prototype and doing a fitting until the prototype is perfectly aligned with the vision

Pattern drawing image: https://www.craftsy.com/post/tips-for-pattern-drafting
Source concept and design images:
https://www.behance.net/gallery/98053429/Fashion-design-portfolio





Fitting → Final patterns → Sewing → Finishing

The Full Pipeline

After fitting, when the prototype has been approved, comes making the final patterns
And then sewing the final garments
And then finishing off with a photoshoot and probably a catwalk show to present the collection

Sewing image: https://www.familyhandyman.com/project/how-to-sew/
Source final finishing image: https://www.behance.net/gallery/98053429/Fashion-design-portfolio

Design Routes Context - fashion

Runway / Fashion Shows

Haute couture

- Made by specialised tailors
- Best quality fabrics
- Unique and exquisite pieces
- Made to measure
- Limited items
- EXPENSIVE

Ready-to-wear (prêt-à-porter)

- Mass produced in large factories
- Standard sizing
- Affordable
- Following trends

I now want to talk to you about the different design routes that can happen in fashion
First is designing for runway/fashion shows.
There are two options for this: haute couture and pret-a-porter.
Haute Couture is made by specialist tailores, the best quality and unique pieces, and is super expensive
And pret-a-porter is mass produced in large factories, affordable and follows trends

Design Routes Context - fashion

Market-driven

Fast fashion

- Cheap
- Trendy
- Made quickly
- Mass produced
- Unethical labour
- Large footprint

Brand examples:
- Shein
- H&M
- Zara
- Primark
- Bershka
- Uniqlo

Slow fashion

- Sustainable
- Environmentally friendly
- Quality
- Durable
- Ethically
- Timeless pieces
- Natural/recycled materials

Brand examples:
- Patagonia
- Levi's
- Stella McCartney
- Kings of Indigo
- Etsy
- Arket

Designing for market-driven fashion is different, there are two options here: fast fashion and slow fashion
Where fast fashions only focus is money, clothing needs to be made as fast and cheap as possible, which is often unethical
And slow fashion focusses on sustainable production, quality and is more ethical for the world and its workers

Design Routes Context - fashion

Conceptual / independent

- Artistic exploration
- Skill development
- Personal identity
- Raise awareness
- Challenge world views
- Heritage/cultural exploration
- Innovation
- New materials / construction

Then there is the conceptual/independent designer, my study trained me to be this kind of designer
Here the focus lays on artistic expression, improving skills and the designers personal identity

•Source image: ruben jurrien: https://www.parool.nl/ps/modeprijswinnaar-ruben-jurrien-ontwerpt-voor-alle-genders-doe-eens-een-jurk-aan-als-man-kijk-hoe-het-voelt~bbf88207/

Design Routes Context - costuming

Costumes for theatre

- Communicate from a distance.
- Colours & materials must hold up under theatrical lighting.
- Exaggerated silhouettes, bold colour contrast.
- Designed for repeat performances + sweat, dance, movement.
- Quick changes → hidden zippers, magnets, rigging.
- Durable for years of theatre.

And two other examples of designing garments is first for costuming
Here the focus lies on communication from a distance while the garments are made to be very durable for years of performance, movement, sweat etc.

Source image shakespear https://www.rsc.org.uk/blogs/out-of-the-spotlight/mens-costume

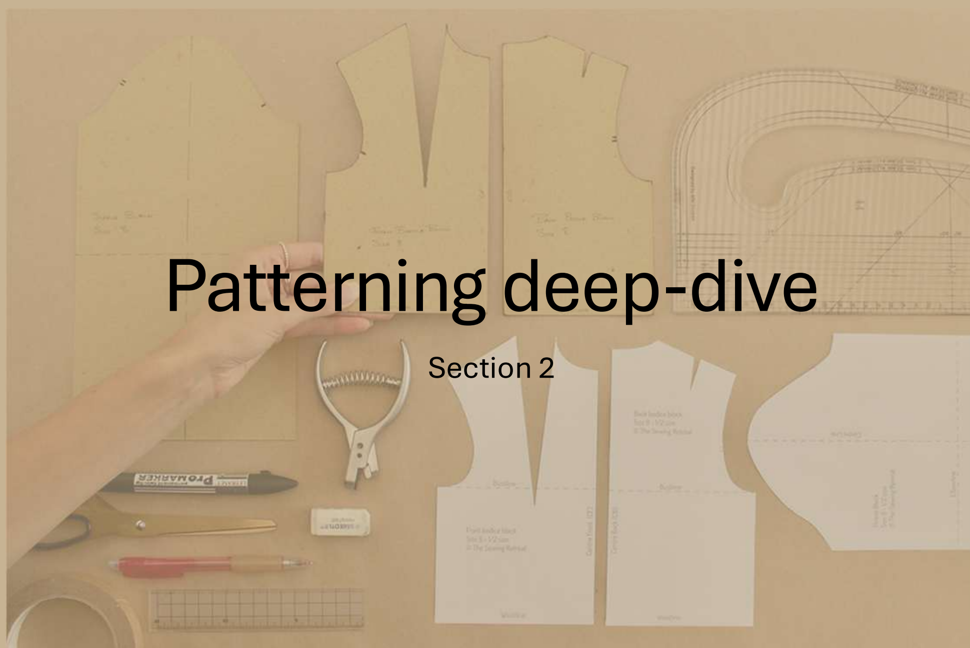
Design Routes Context - costuming

Costumes for film

- Costumes must look real on camera.
- More subtle, nuanced design.
- Close-up detailing is essential.
- Multiple versions needed for stunts, damage, continuity.
- Often exists just for the scene.

And finally designing for film, which focusses more on looking real on camera, especially close up shots.
These garments might just be worn only once, just for a scene and then disposed of.

Source image: https://theatreandfilm.capetown/iconic-movie-costumes-character-identities/



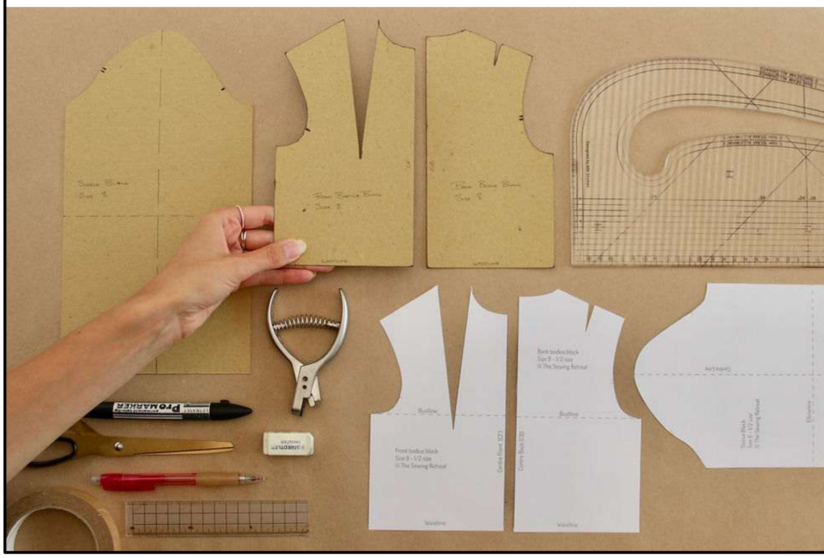
Patterning deep-dive

Section 2

Let’s dive into patterning

Image source: https://www.thesewingretreat.co.uk/post/what-do-you-need-to-start-designing-your-own-sewing-patterns

What is Patterning?



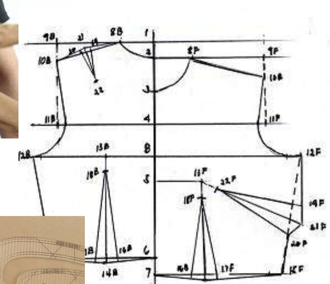
Definition:
Creating flat templates that, when cut from fabric and sewn, create 3D garments

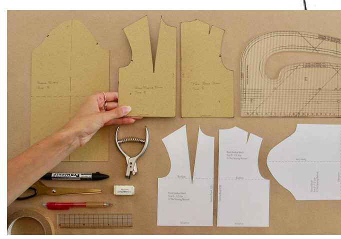
What is patterning?
The Definition: Creating flat templates that, when cut from fabric and sewn, create 3D garments
You can think of creating patterns as uv unwrapping a persons body, just with some more space added in shape of fabric

Traditional Patterning Methods

Method 1: Flat Pattern Drafting

1. Take measurements
2. Calculate dimensions
3. Draft basic shapes (geometry)
4. Add shaping
5. Trace master pattern
6. Add symbols and labels
7. Ready for prototyping





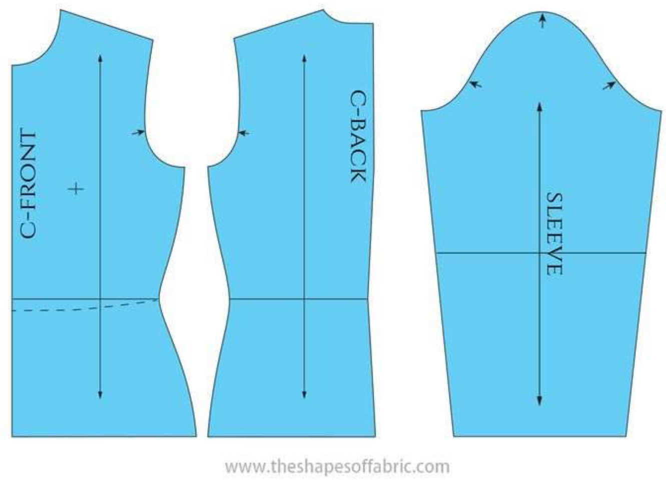
The first traditional patterning method is flat pattern drafting. This one takes the longest and is very reliant on math

- Start with body measurements
- Mathematical calculations and geometry
- Draw directly onto large pattern paper
- Requires understanding of ease, grain, proportions

Traditional Patterning Methods

Method 2: Basic blocks

1. Start from basic block
2. Edit to desired design
3. Trace and cut
4. Ready for prototyping



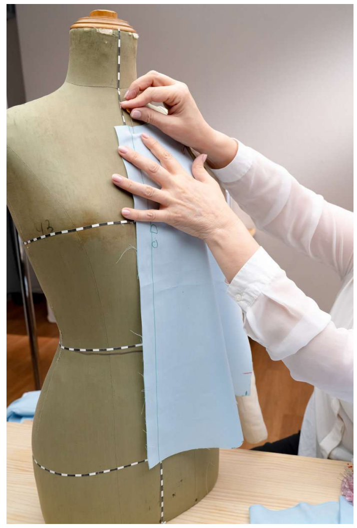
Basic fitted templates for torso, sleeve, skirt, pants
Starting point that gets modified for different designs
Like having a base mesh in 3D that you edit to your needs

Traditional Patterning Methods

Method 3: Draping

1. Optional: map out design
2. Pin fabric to dress form
3. Manipulate fabric
4. Create desired shape
5. Take off dress form
6. Mark and trace
7. Create flat patterns
8. Ready for prototyping

Similar to asset sculpting



This is a video, hover and play

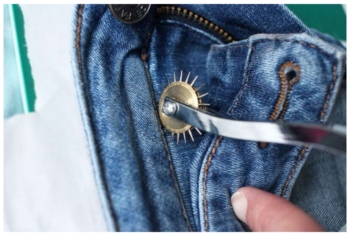
Then this method, draping, is more intuitive, almost like sculpting in 3d

Pin fabric directly onto dress form
Manipulate fabric to create desired shape
Mark and trace the result into flat patterns
More intuitive, sculptural approach

Traditional Patterning Methods

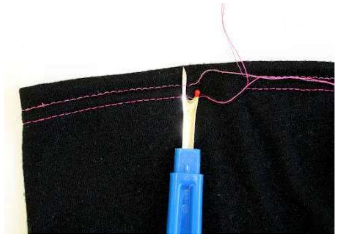
Method 4: Tracing

1. Buy something
2. Trace it
3. Smooth and straighten lines
4. Trace again
5. Ready for prototype

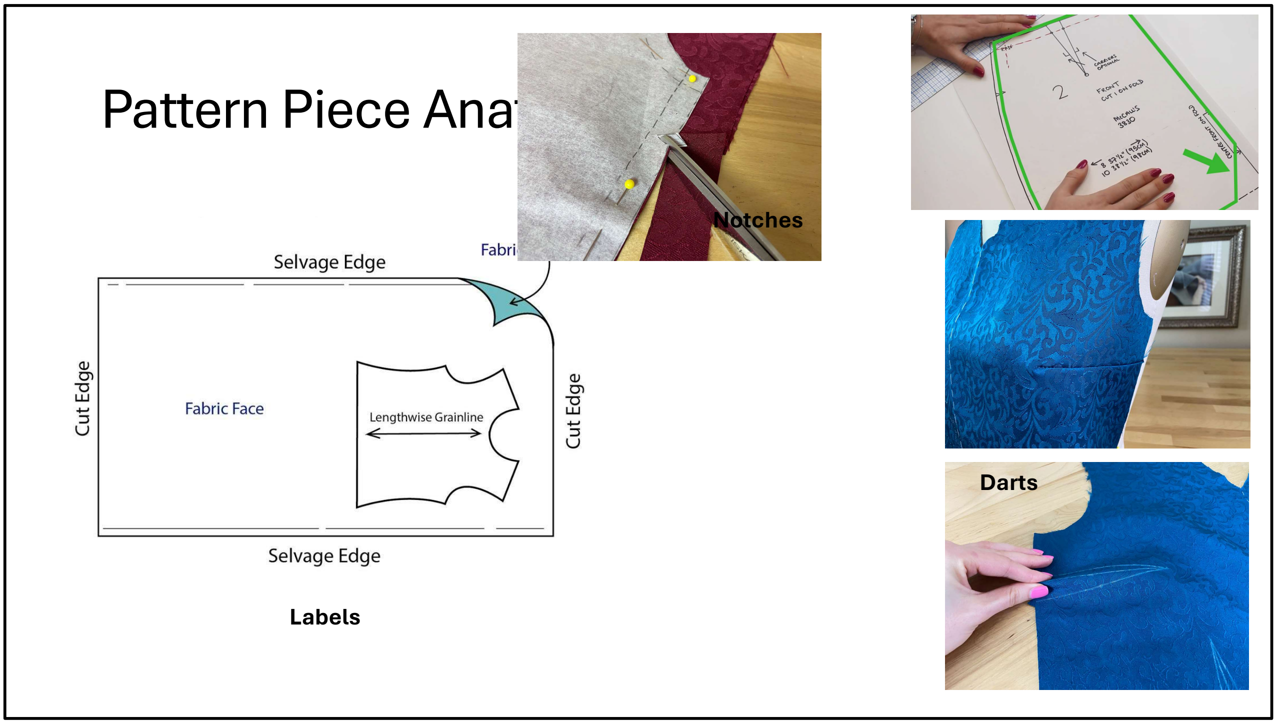


Method 5: Deconstructing

1. Buy something
2. Buy it again (vinted)
3. Take it apart
4. Trace patterns
5. Ready for prototype



And the two final methods are using existing garments
First tracing, in which you use a spiky pizza knife kind of tool to trace the patterns onto paper by punching holes in it without damaging the garment
And the second is deconstruction, where you take apart a garment and trace the patterns that way



- Grain line (direction matters!)
- Notches (matching points)
- Darts (that's how you add 3D shaping to flat fabric)
- Seam allowance (the edge you sew together)
- Labels (front bodice, back bodice, etc. so you know which part is what and what it creates)

Image source lines: https://www.doinaalexei.com/beginnersewingtutorialblog/4-ways-to-lay-out-a-sewing-pattern-for-cutting-understanding-fabric-grain-and-the-selvage-edge
Notches: https://www.doinaalexei.com/what-are-notches-and-how-to-use-them-in-the-sewing-process.html
Darts: https://www.doinaalexei.com/what-are-darts-and-dart-components.html
Labels: https://www.wikihow.com/Sew-Using-Patterns

Activity!

The pattern puzzle

Time for an activity, the pattern puzzle

Wall of pattern pieces image source: https://pin.it/2tYn0XKXw

Activity timeline

Setup (2 min)

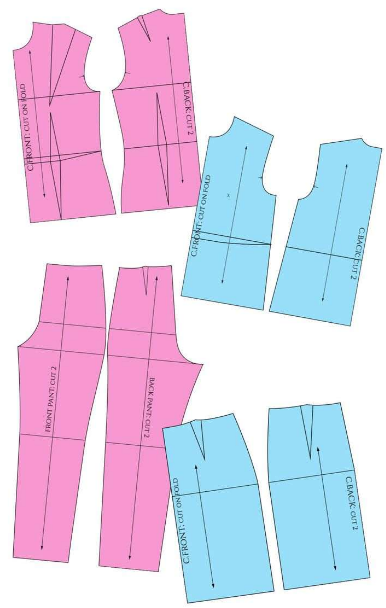
- Divide into pairs/small groups
- Each group gets envelope with pattern pieces
- Brief

Activity Time (8 min)

- Figure out what garment you're holding
- Create the construction layout

Group Share (3 min)

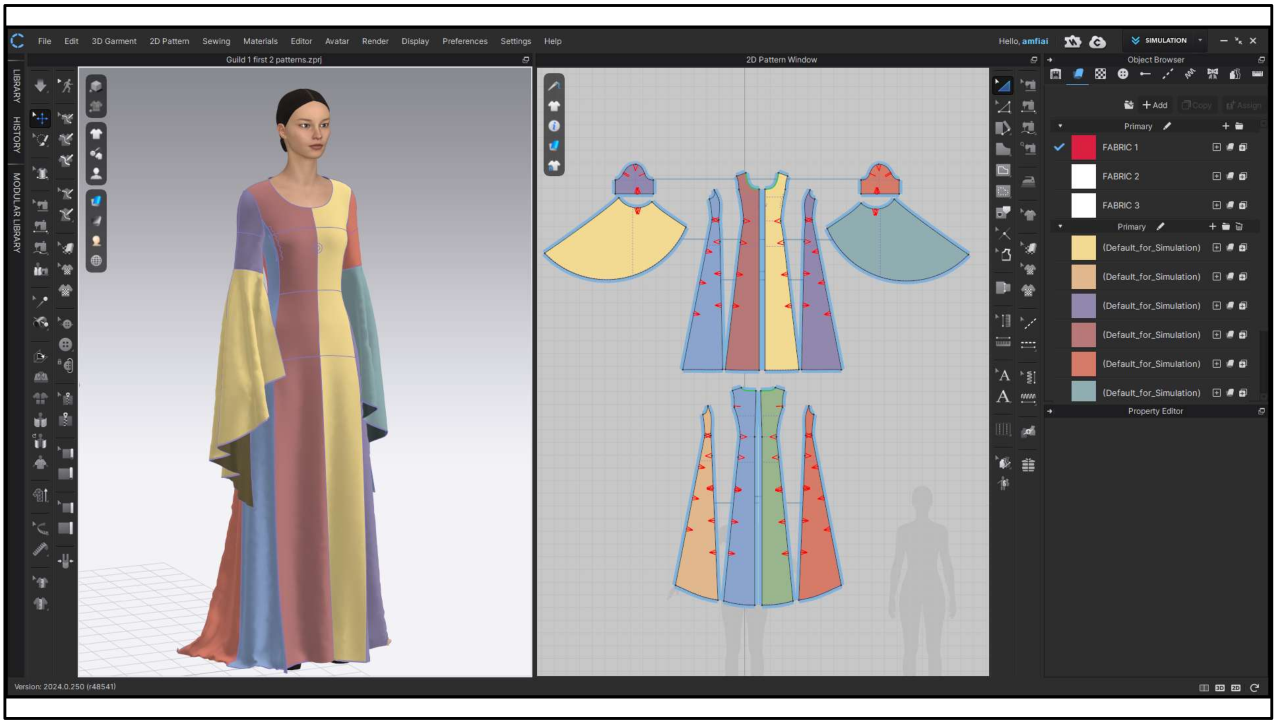
- What did you figure out?
- What clues helped?
- What was confusing?



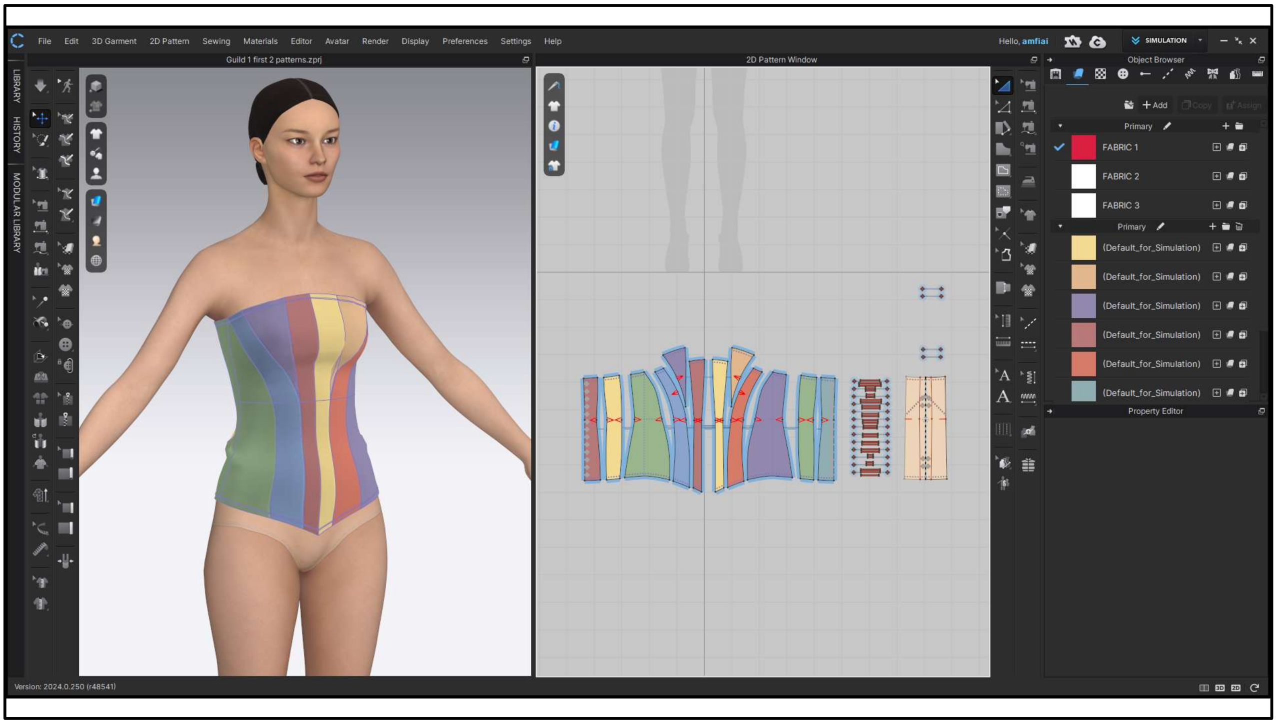
Setup (2 min)
Divide into pairs/small groups
Each group gets envelope with pattern pieces
Figure out what garment you're holding

Activity Time (8 min) Students work on puzzle
You circulate, observe, ask guiding questions
"What makes you think it's X?"
"Where would this piece go on the body?"

Group Share (3 min)
Quick go-around: what did you figure out?
What clues helped? (notches matching, obvious shapes like crotch curve for pants)
What was confusing? (lining pieces, cowl draping, pieces that don't look like body parts)



Medieval dress with long sleeves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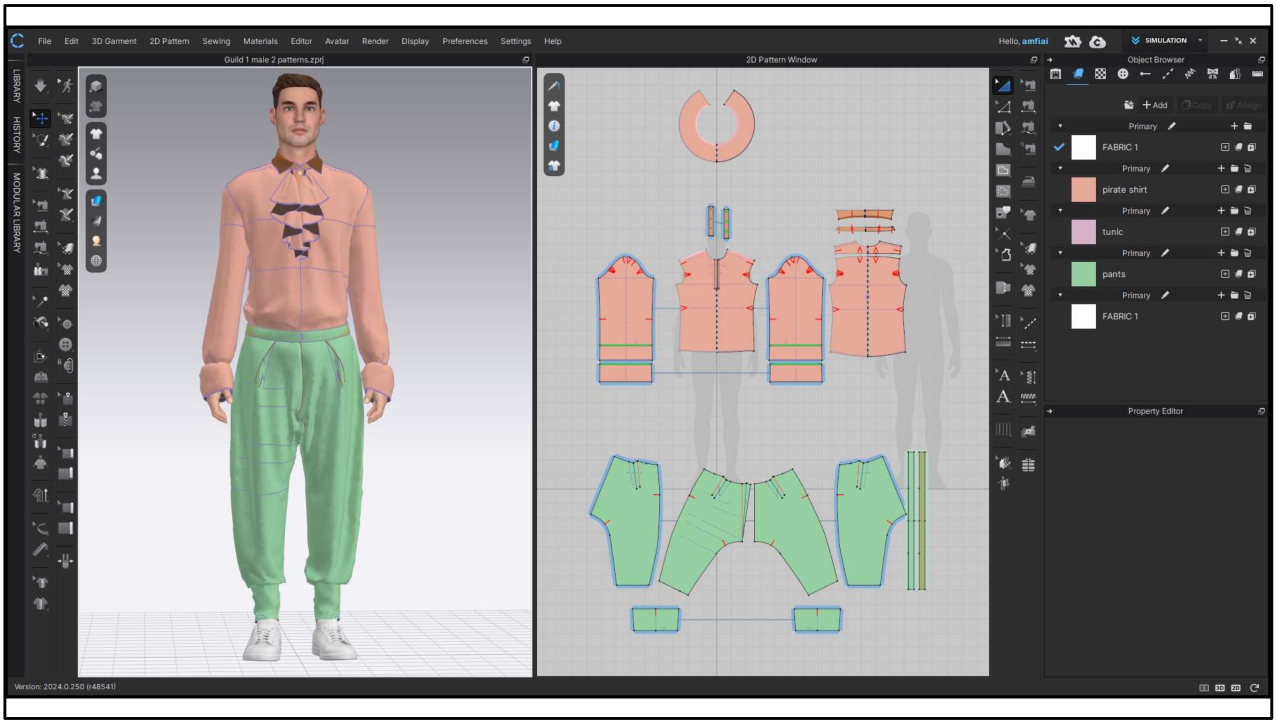
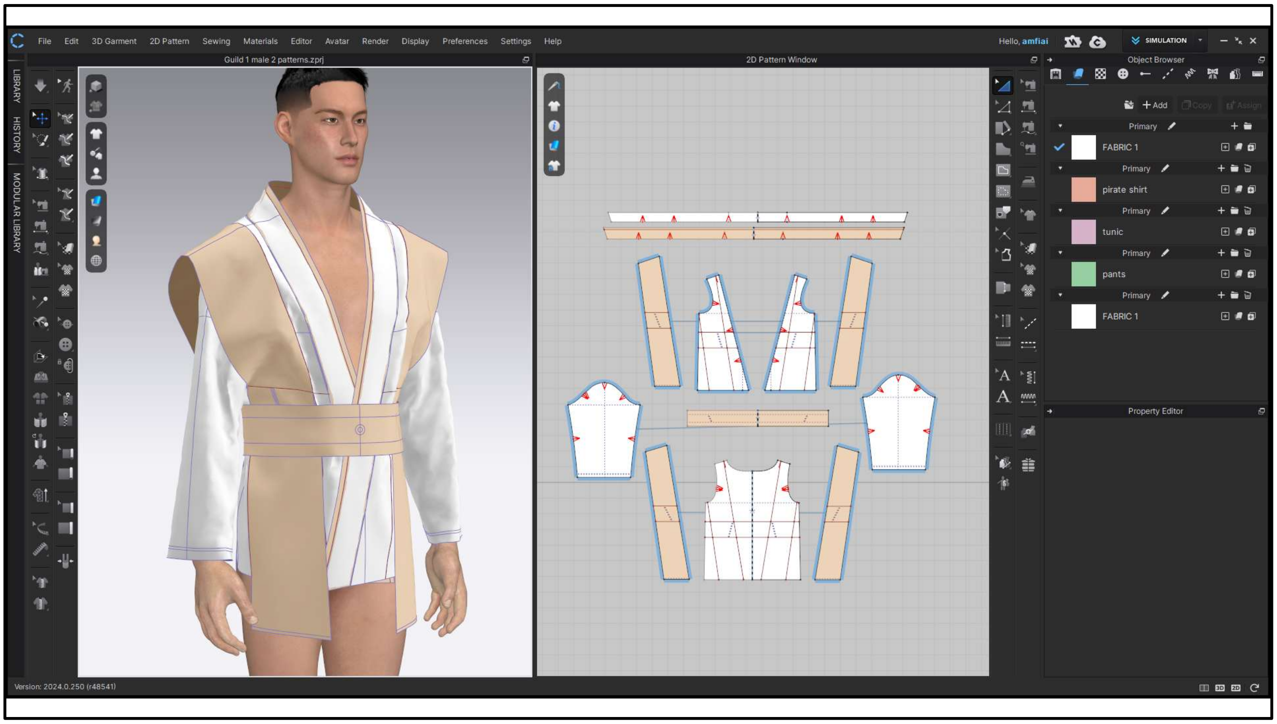
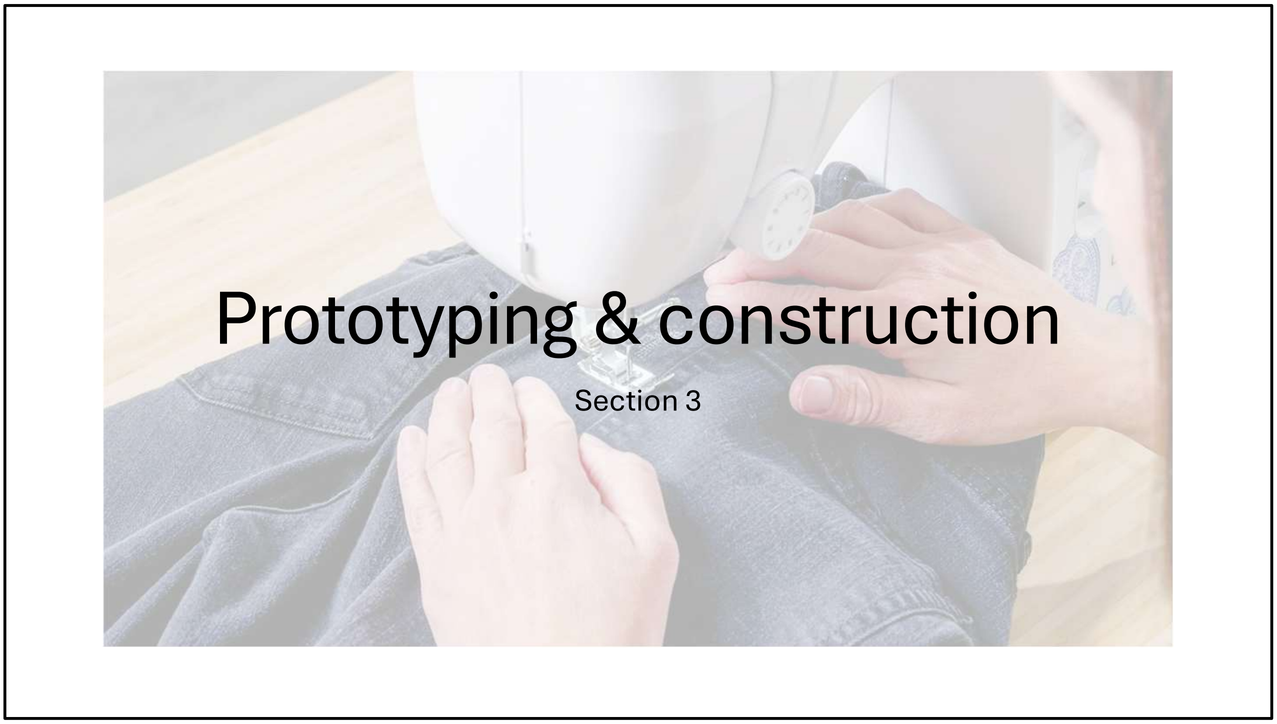
corset



Pirate shirt with volant at the front
Wide pants with calf cuffs



And this jedi-looking jacket thing



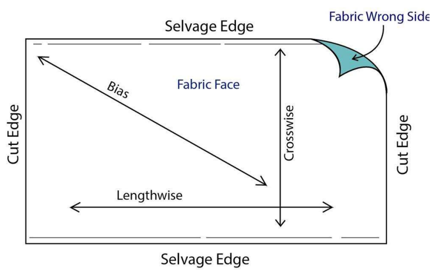
Now lets talk about prototyping and construction

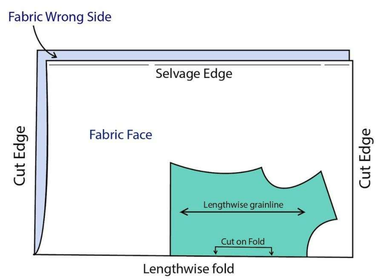
Image source: https://www.familyhandyman.com/project/how-to-sew/

From Patterns to Prototype

Fabric Preparation & Cutting

- Laying out patterns on fabric
- Cutting with seam allowance
- Marking notches and important points





When going from patterns to prototype you need to cut them out of fabric first
Making sure you keep to the "rules" of the fabric: grainline, right side/wrong side, on fold or not etc
Make sure you mark notches, so you can align them while sewing so you know its in the right place

Source fabric anatomy & piece on fabric image: https://www.doinaalexei.com/beginnersewingtutorialblog/4-ways-to-lay-out-a-sewing-pattern-for-cutting-understanding-fabric-grain-and-the-selvage-edge
Pieces on fabric image: https://houseofmisssew.com/anthea-blouse-anna-allen-patterns

From Patterns to Prototype

Sample Making

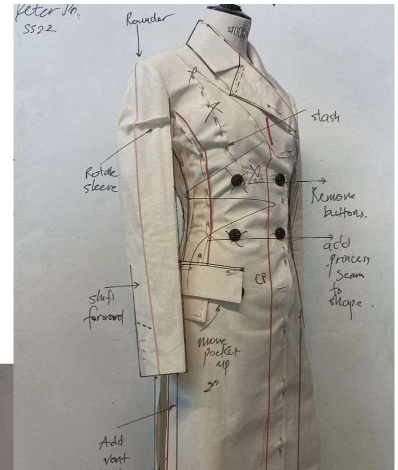
- First version in cheap fabric (muslin/calico)
- Called a "toile" or "muslin"
- Test if your patterns actually work

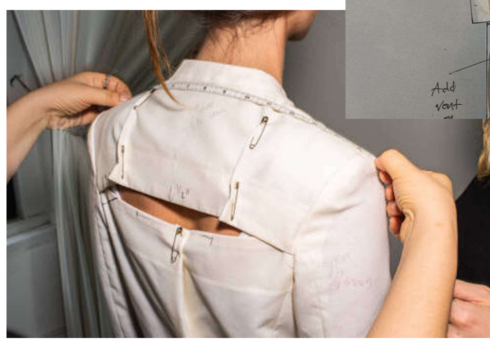
First version usually in cheap fabric (muslin/calico)
Called a "toile" or "muslin"
This is where you test if your patterns actually work

Source all 3 images: https://www.allsewnupwales.co.uk/product/calico-for-sampling/

The Fitting Process

1. Put sample garment on person
2. Check fit
3. Mark adjustments
4. Cut into it
5. Transfer changes back to patterns
6. Make new sample
7. Repeat until perfect





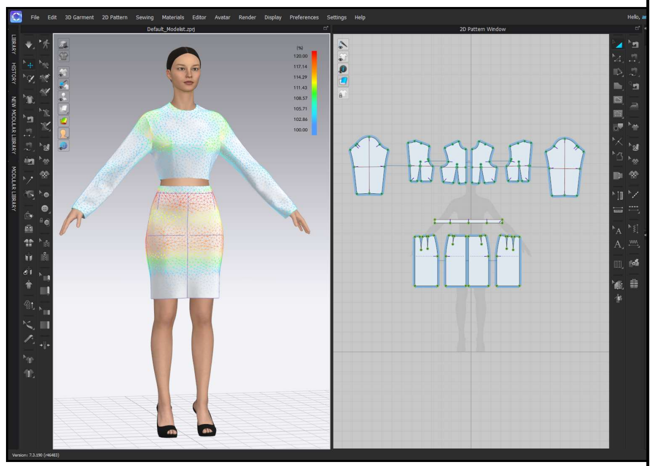
Put sample on dress form or live model
Check: Does it fit? Does it move right? Is the silhouette correct?
Mark adjustments directly on the sample
Common issues: too tight, too loose, wrinkles/pulling, proportions off
Transfer adjustments back to patterns
Make new sample
Repeat until perfect , which sometimes takes 10+ samples which is a bit of a waste

Source fitting image https://www.vmora.com/blog-and-press/steps-to-production-fitting

Digital Prototyping with CLO3D

The Modern Workflow

1. Import or create patterns digitally
2. Place on avatar with correct measurements
3. Run physics simulation
4. See immediately: fit issues, drape
5. Make adjustments in real-time
6. Saves material and time
7. BUT: still need at least one physical sample



3D scanning yourself, designing outfit digitally
Is supercool for cosplay/larping btw

But The Modern Workflow fixes a part of this wasteful process:
You can do fittings digitally, already fix issues that could take you a few fittings and prototypes as you can see your garment in 3D already
And if you have your model person 3D scanned (or yourself) you have the right measurements already
With tools like the strain map you can see where there's not enough space and you can fix it before you even touched the sewing machine

Process:
Import or create patterns digitally
Place on avatar with correct measurements
Run physics simulation
See immediately: fit issues, drape, how fabric behaves
Make adjustments in real-time
Saves material and time (fewer physical samples needed)
BUT: still need at least one physical sample (fabric behaves differently than simulation)

Even with perfect patterns, construction has challenges
Sewing order matters (can't sew this seam after that one is closed)
Fabric properties affect outcome (stretch vs. woven, drape, weight)
Details like interfacing, linings, closures add complexity
Finishing: hems, topstitching, pressing/ironing
This is why couture garments are expensive - hours of skilled hand work

(I showed an example of a blouse I sewed myself: took 10 hours of work, and if you pay me below minimum wage,
it would still cost 100+ euros to buy this garment and then it’s only man-work hour and not yet materials or fair wages.)

Blue garment image source: https://www.thesprucecrafts.com/free-patterns-to-sew-clothing-2977357
ironing source image: https://jinfengapparel.com/why-is-ironing-so-important-in-sewing/
sewing image: https://www.popsci.com/story/diy/sewing-repair-alteration-tips/
Topstitching image: https://www.thelaststitch.com/how-to-sew-perfect-topstitching/

Believable costume silhouettes come from understanding construction
You'll know: where seams go, how fabric drapes, what's physically possible
Better communication with 3D artists who model your designs
You can make intentional choices about when to break reality

•Source: https://www.artstation.com/zahraa97

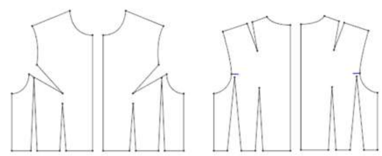
Time for the second activity, creating 3d forms

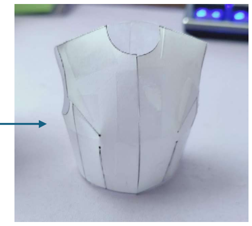
Image source: https://www.refinery29.com/en-us/posadas-origami-paper-garments-are-the-most-hazardous-clothes-ever

Activity timeline

Setup (2 min)

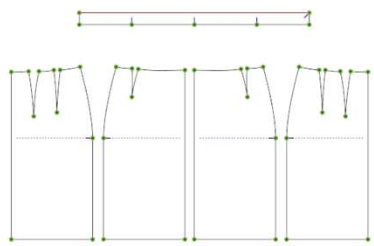
- Choose a pattern

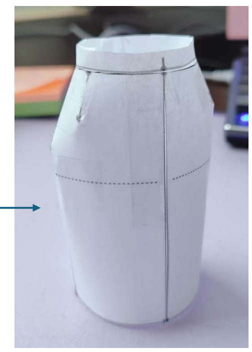




Activity Time (8-10 min)

- Cut the patterns
- Only cut 1 line of the darts for easy taping
- Tape the mini garment together





Debrief (2 min)

- What did you discover?
- Did you find it easy/hard?

Setup (2 min)
"Remember UV unwrapping? Let's do it in real life"
Pairs get: object (sphere/cube/pyramid/cylinder) + painters tape + scissors + paper
Challenge: Cover your object completely in tape, figure out where to cut it to make it lay flat, trace those shapes - those are your patterns

Activity Time (8-10 min)
Students experiment with covering objects
They discover: spheres need multiple pieces/darts, cubes are easier, curves are tricky
They make decisions about seam placement
Remove tape, flatten, trace patterns
If time/interest: swap objects and try another

Debrief (2 min)
What did you discover?
Which shapes were easy vs. hard?
"This is exactly what pattern makers do with the human body - but bodies are way more complex than spheres"
"Draping on a dress form is like what you just did - cover the 3D form, mark where to cut, flatten into patterns"

POST-SESSION QUESTIONS

Red:
- No
- No experience
- Unfamiliar
- Not confident

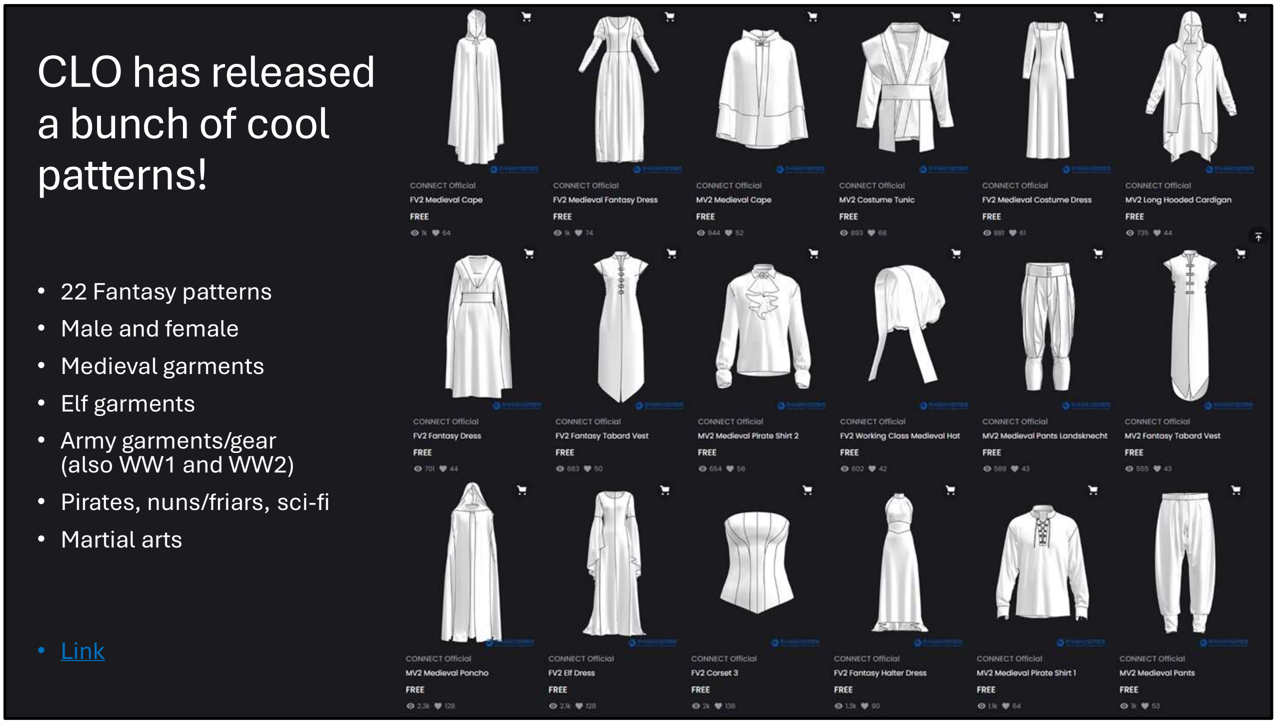
Yellow:
- Maybe
- Some experience
- Somewhat familiar
- Moderately confident

Green:
- Yes
- Experienced
- Very familiar
- Very confident

Post-Assessment with Colored Cards Same questions as pre-assessment:
- Have you ever sewn a garment before? (shouldn't change)
- Can you identify what garment a flat pattern makes just by looking at it?
- Do you know what a basic block/sloper is?
- How confident do you feel understanding how flat patterns become 3D garments? (scale)
- Have you used any digital fashion software? (shouldn't change)

Plus possibly:
- Do you feel this knowledge would help you design more believable character costumes?
- Would you be interested in learning more about garment construction?

Thanks for joining and I hope to see you next week!
Where we will cover: fabrics, materials and natural dying
Natural dying is especially nice to dive into as this could be used to tell stories about geolocation of your characters
For example if they live in a place with clay, they might dye with it and wear grey/black clothing, or a space with snails, could become purple clothing etc



And here a nice resource, clo (the maker of marvelous and clo3d) has recently released a bunch of awesome basic blocks
All free to download and use for your projects, making the process easier and faster for you.